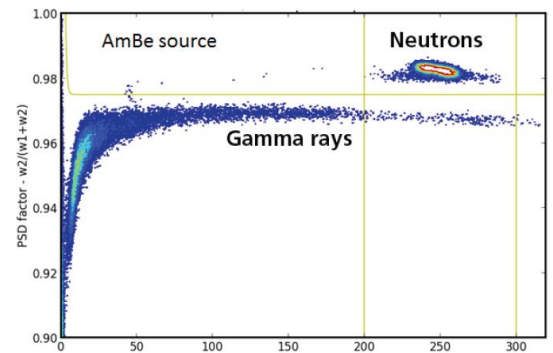


CLYC:Ce

Description: $\text{Cs}_2\text{LiYCl}_6:\text{Ce}$ Scintillation crystals are used for neutron and gamma detection. Because of the neutron interaction with Cl-35, CLYC can also be used for fast neutron detection.

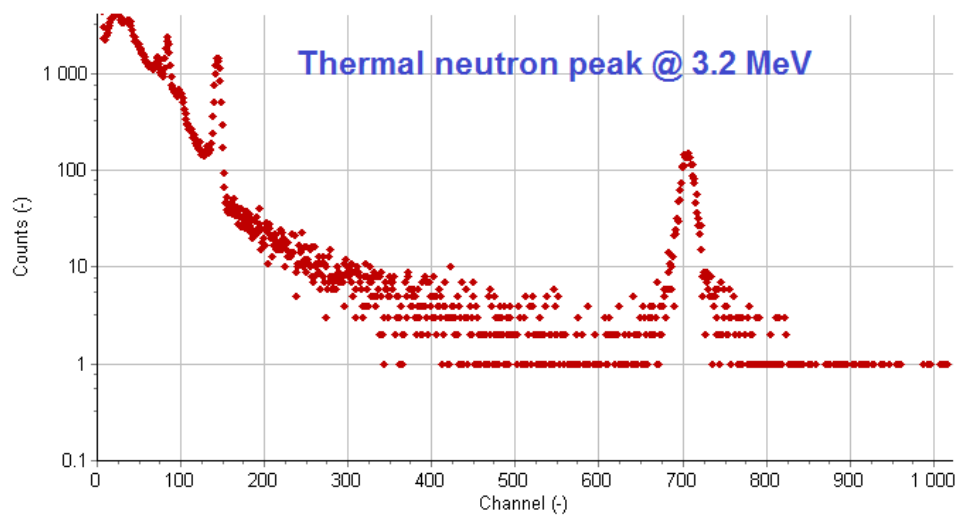
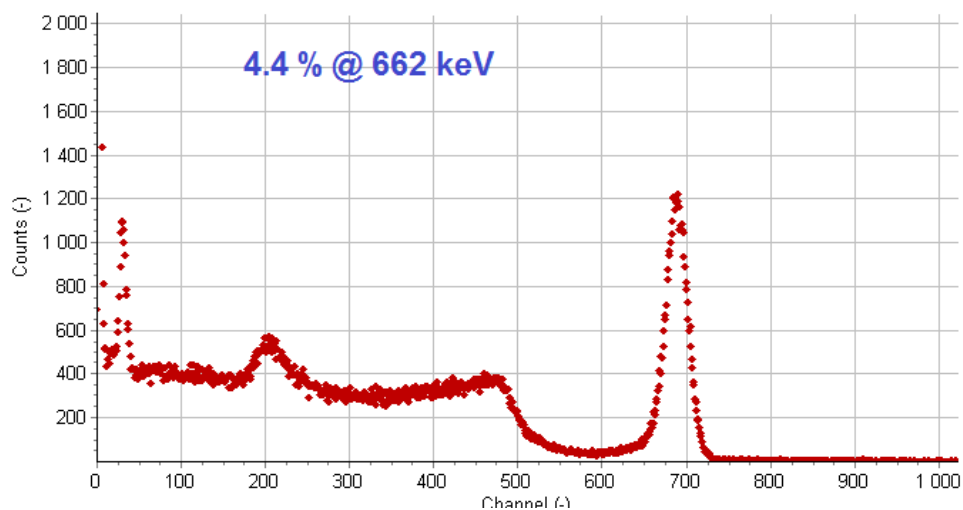
Scintillation crystal	:	CLYC (Cerium doped Caesium Lithium Yttrium Chloride)
Melting point	:	640 °C
Hygroscopic	:	Yes
Li-6 enrichment	:	96%
Density	:	3.31 g/cm ³
Refractive index	:	1.81 (400 nm)
Wavelength of Emission	:	275 - 450 nm
Max emission wavelength	:	370 nm
Decay time constants	:	1 ns, 50 ns, 1000 ns
Scintillation light yield	:	approx.. 20.000 photons/MeV
Pulse shape discrimination	:	Well possible to discriminate neutrons from gammas



Performance

Energy resolution (662 keV)	:	4.5 – 5.5% FWHM depending on geometry
Thermal Neutron peak	:	3.1 – 3.3 MeV

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